

Problem

Simplify the following:

(a) $f(t) = 5 \cos(2t + 15^\circ) - 4 \sin(2t - 30^\circ)$

(b) $g(t) = 8 \sin t + 4 \cos(t + 50^\circ)$

(c) $h(t) = \int_0^t (10 \cos 40t + 50 \sin 40t) dt$

Step-by-step solution

Step 1 of 3

(a)

Consider the following expression:

$$\begin{aligned} f(t) &= 5 \cos(2t + 15^\circ) - 4 \sin(2t - 30^\circ) \\ &= 5 \cos(2t + 15^\circ) - 4 \cos(2t - 120^\circ) \end{aligned}$$

Convert into polar form.

$$\begin{aligned} f(t) &= 5 \angle 15^\circ - 4 \angle (-120^\circ) \\ &= (4.829 + j1.294) - (-2 - j3.464) \\ &= 4.829 + j1.294 + 2 + j3.464 \\ &= 6.829 + j4.758 \\ &= 8.323 \angle 34.86^\circ \end{aligned}$$

Therefore, the final expression for $f(t)$ is,

$$\boxed{f(t) = 8.323 \cos(2t + 34.86^\circ)}.$$

Step 2 of 3

(b)

Consider the following expression:

$$\begin{aligned} g(t) &= 8 \sin t + 4 \cos(t + 50^\circ) \\ &= 8 \cos(t - 90^\circ) + 4 \cos(t + 50^\circ) \end{aligned}$$

Convert into polar form.

$$\begin{aligned} g(t) &= 8 \angle -90^\circ + 4 \angle 50^\circ \\ &= (-j8) + (2.571 + j3.064) \\ &= 2.571 - j4.935 \\ &= 5.565 \angle -62.48^\circ \end{aligned}$$

Therefore, the final expression for $g(t)$ is $\boxed{g(t) = 5.565 \cos(t - 62.48^\circ)}.$

Step 3 of 3

(c)

Consider the following expression:

$$\begin{aligned}h(t) &= \int_0^t (10 \times \cos 40t + 50 \times \sin 40t) dt \\&= \int_0^t (10 \times \cos 40t) dt + \int_0^t (50 \times \sin 40t) dt \\&= \int_0^t 10 \cos 40t dt + \int_0^t 50 \cos(40t - 90^\circ) dt \\&= \frac{10}{40} \sin 40t + \frac{50}{40} \sin(40t - 90^\circ) \\&= 0.25 \sin 40t + 1.25 \sin(40t - 90^\circ) \\&= 0.25 \cos(40t - 90^\circ) - 1.25 \cos(40t)\end{aligned}$$

Convert into polar form.

$$\begin{aligned}h(t) &= 0.25 \angle -90^\circ - 1.25 \angle 0^\circ \\&= -1.25 - j0.25 \\&= 1.2748 \angle -168.69^\circ\end{aligned}$$

Therefore, the final expression for $h(t)$ is $1.2748 \angle -168.69^\circ$.